

DuoSubaccountUserLimit

Disclaimer: This is an unofficial tool and is not supported by Cisco or Duo Security. It relies on an undocumented API endpoint that may change or be removed without notice. Use at your own risk.

PowerShell module for MSP admins to list Duo child accounts and get/set hard user limits via the undocumented `/admin/v1/billing/user_limit` endpoint.

Requirements

- PowerShell 7+
- Duo **Accounts API** credentials (not Admin API) from your parent MSP account

Setup

Set the following environment variables before importing the module:

```
$env:DUO_ACCOUNTS_IKEY = 'your-accounts-integration-key'  
$env:DUO_ACCOUNTS_SKEY = 'your-accounts-secret-key'  
$env:DUO_ACCOUNTS_HOST = 'api-accounts.duosecurity.com'
```

Import

```
Import-Module ./duo-subaccount-hard-user-limit/DuoSubaccountUserLimit.psd1
```

Usage

List all child accounts

```
Get-DuoChildAccounts
```

Returns objects with `name`, `account_id`, `api_hostname`, and other fields from the Accounts API.

Get user limit for a specific account

```
Get-DuoUserLimit -AccountId DA12345 `  
-ApiHostname api-XXXXXXX.duosecurity.com
```

Get user limits for all child accounts

```
Get-DuoChildAccounts | Get-DuoUserLimit
```

Set a user limit

```
Set-DuoUserLimit -AccountId DA12345 `  
-ApiHostname api-XXXXXXX.duosecurity.com `  
-UserLimit 100
```

Set a limit by account name (using pipeline)

```
Get-DuoChildAccounts |  
    Where-Object name -eq 'Acme Corp' |  
    Set-DuoUserLimit -UserLimit 50
```

Remove a limit (set to 0)

```
Set-DuoUserLimit -AccountId DA12345 `  
    -ApiHostname api-XXXXXXX.duosecurity.com `  
    -UserLimit 0
```

Preview a set operation without applying it (-WhatIf)

```
Get-DuoChildAccounts |  
    Where-Object name -eq 'Acme Corp' |  
    Set-DuoUserLimit -UserLimit 50 -WhatIf
```

Bulk report: all accounts with their current limits

```
Get-DuoChildAccounts | Get-DuoUserLimit | Format-Table
```

Example walkthrough

A full session: list accounts, check limits, set limits on a couple, then remove one.

1. Import the module

```
Import-Module /path/to/duo-subaccount-hard-user-limit/DuoSubaccountUserLimit.psd1
```

2. Set credentials

```
$env:DUO_ACCOUNTS_IKEY = 'DIXXXXXXXXXXXXXXXXXXX'  
$env:DUO_ACCOUNTS_SKEY = 'your-accounts-secret-key'  
$env:DUO_ACCOUNTS_HOST = 'api-accounts.duosecurity.com'
```

3. List all child accounts

```
Get-DuoChildAccounts
```

<i># account_id</i>	<i>api_hostname</i>	<i>name</i>
<i># -----</i>	<i>-----</i>	<i>----</i>
<i># DAXXXXXXXXXXXXXXXXXXX</i>	<i>api-XXXXXXX.duosecurity.com</i>	<i>Acme Corp</i>
<i># DAYYYYYYYYYYYYYYYYYYY</i>	<i>api-YYYYYYY.duosecurity.com</i>	<i>Globex Corporation</i>
<i># DAZZZZZZZZZZZZZZZZZZZ</i>	<i>api-ZZZZZZZ.duosecurity.com</i>	<i>Initech</i>

4. Check current limits across all accounts

```
Get-DuoChildAccounts | Get-DuoUserLimit | Format-Table
```

<i># account_id</i>	<i>user_limit</i>
<i># -----</i>	<i>-----</i>
<i># DAXXXXXXXXXXXXXXXXXXX</i>	<i>Not set</i>

```

# DAYYYYYYYYYYYYYYYYYYYY Not set
# DAZZZZZZZZZZZZZZZZZZZ Not set

# 5. Set a limit of 50 on Acme Corp
Get-DuoChildAccounts |
    Where-Object name -eq 'Acme Corp' |
    Set-DuoUserLimit -UserLimit 50

# 6. Set a limit of 100 on Globex Corporation
Get-DuoChildAccounts |
    Where-Object name -eq 'Globex Corporation' |
    Set-DuoUserLimit -UserLimit 100

# 7. Verify the limits were applied
Get-DuoChildAccounts | Get-DuoUserLimit | Format-Table

# account_id          user_limit
# -----
# DAXXXXXXXXXXXXXXXXXX 50
# DAYYYYYYYYYYYYYYYYYY 100
# DAZZZZZZZZZZZZZZZZZZ Not set

# 8. Remove the limit from Acme Corp (set to 0)
Set-DuoUserLimit -AccountId 'DAXXXXXXXXXXXXXXXXXX' `
    -ApiHostname 'api-XXXXXXX.duosecurity.com' `
    -UserLimit 0

# 9. Confirm it was removed
Get-DuoUserLimit -AccountId 'DAXXXXXXXXXXXXXXXXXX' `
    -ApiHostname 'api-XXXXXXX.duosecurity.com'

# account_id          user_limit
# -----
# DAXXXXXXXXXXXXXXXXXX Not set

```

How it works

Authentication

The module uses the **Accounts API credentials** with the **child account's API hostname** — the standard Duo MSP authentication pattern:

Accounts IKEY + Accounts SKEY → signed against child account's hostname

This is the same pattern used in the Python scripts in this repo.

Signing

Duo uses HMAC-SHA1 over a canonical string of:

date / METHOD / host / path / params

This is handled internally and not exposed as a callable function.

Module structure

```
duo-subaccount-hard-user-limit/  
  DuoSubaccountUserLimit.psd1      Module manifest  
  DuoSubaccountUserLimit.psm1      Module loader  
  Private/  
    Invoke-DuoSignedRequest.ps1    HMAC-SHA1 signing + HTTP (not exported)  
    Get-DuoCredentials.ps1         Reads env vars (not exported)  
  Public/  
    Get-DuoChildAccounts.ps1  
    Get-DuoUserLimit.ps1  
    Set-DuoUserLimit.ps1
```